

Determining the long-term dynamical and temperature stability of potentially habitable planets in trinary star systems GJ 667 and LTT 1445

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ABSTRACT

Aims. Investigate long-term dynamical and temperature stability on exoplanets of systems Gliese 667, LTT 1445, HD 132563, Kepler-444, and 94 Ceti.

Methods. Used a Wisdom-Holman integrator adapted for triple hierarchical star systems and temperature tracking. For systems without orbital parameters for stellar orbits, explored a range of eccentricities and inclinations.

Results. 2 out of 2 habitable exoplanets in Gliese 667 remain stable long term, 94 Ceti and Kepler 444 support the existence of synthetic exoplanets, while LTT 1445 is highly dynamically unstable and HD 132563 has very high temperature changes.

Conclusions. With this work long-term stability can be determined, but not necessarily habitability by only using the habitable zone and temperature estimates.

Key words. exoplanets – N-body simulation – habitability

1. Introduction

The search for potentially habitable planets has always been a core part of astrophysics. From the first exoplanet discovered in 1992 around pulsar PSR B1257+12 (Wolszczan & Frail 1992) to today, more than 6000 exoplanets have been listed in the NASA Exoplanet Archive (NASA Exoplanet Archive 2026). A non-negligible fraction of known exoplanets orbit stars that are members of binary or higher-order multiple systems, although the inferred fraction depends on the sample and observa-

tional methods (e.g. Roell et al. 2012; Michel & Mugrauer 2021; Schlagenhauf et al. 2024; Raghavan et al. 2006). Due to the complex orbital architecture of multiple star systems and their exoplanets, analyzing the dynamical stability of such systems is crucial for long-term habitability. The presence of other stellar companions can be the driver of the Kozai-Lidov mechanism (Kozai 1962; Lidov 1962), influencing the temperature and climate of a potentially habitable exoplanet. Stable zones in binary systems have been widely studied, with boundaries of stable zones empirically defined for all types of planetary orbits (Holman & Wiegert 1999). Related work studied habitable zones of binary systems (e.g. Haghighipour & Kaltenegger 2013; Kaltenegger & Haghighipour 2013; Eggl et al. 2020). For triple hierarchical systems most body of work focuses on dynamical stability (Verrier & Evans 2007; Busetti et al. 2018), with more recent work exploring habitable zones in such systems (e.g. Müller & Haghighipour 2014). Even though there are only 73 triple star systems listed in the NASA Exoplanet archive (NASA Exoplanet Archive 2026), they offer a wide range of planetary configurations and potential habitable exoplanets. 2 very interesting and close cases, within 24 light years, are systems Gliese 667 and LTT 1445. Gliese 667 C hosts 2 Super-Earths, with GJ 667 Cc being in the habitable zone and an exciting candidate for habitability (Anglada-Escudé et al. 2012; Delfosse, X. et al. 2013), although there has been significant debate over the total number of planets. (Anglada-Escudé, Guillem et al. 2013) claimed there are 6, or even 7 planets in the system with 3 of them being in the habitable zone, while later research (F. Feroz 2014; Robertson & Mahadevan 2014) argues that only GJ 667 Cb and GJ 667 Cc are confirmed, with the other planets being artifacts on the RV periodograms. While GJ 667 Cc appears to be a prime target for habitability, its closeness to its host M dwarf causes multiple issues affecting habitability, such as rising temperatures due to spin-down time (Makarov & Berghua 2013) and flares of UV and X-rays which can harm any potential life if the planet has no magnetic field (Sloane et al. 2023). LTT 1445 A hosts 2 planets as well, LTT 1445 Ab and LTT 1445 Ac with a possible third candidate, LTT 1445 Ad. (Winters et al. 2019, 2022) Planets LTT 1445 Ac and Ad have potential to be habitable, as LTT 1445 Ac has a similar radius and density as the Earth (Pass et al. 2023), making it an exciting candidate for further studies, as well as LTT 1445 Ad which is in the habitable zone. However, LTT 1445 Ab is also an interesting candidate for further investigation of its atmosphere, as (Bennett et al. 2025) found traces of HCN and a 2σ certainty of an atmosphere. The rocky planets in this system also face similar issues as in GJ 667, with X rays from the host star LTT 1445 A having the potential to harm water retention, although planet LTT 1445 Ad could still keep its water supply if it contains one. In this work the goal is to investigate the long-term dynamical stability of planets GJ 667 Cb and Cc, and LTT 1445 Ab, Ac and Ad, determine the influence of the trinary system on the dynamics of the planets and the effective temperature, as well as track how the temperature changes as the planets progress in their dynamical evolution using N-body integrators. The data used is presented in Sect. 2, the methodology of an adapted N-body integrator in Sect. 3, the results and discussion in Sect. 4 and Sect. 5, and finally the conclusions of this work in Sect. 6.

2. Data

To investigate the long-term dynamical and temperature stability of potentially habitable exoplanets, orbital and photometric data from all stellar and planetary components is required. Tables 1, 2 and 3 contain all the data used in this work. The luminosities of the stellar components will be used to calculate the effective surface temperature on the exoplanets, further explained in section 3.1. To ensure all the orbital data is in the same reference plane and the planets start at the correct phase, a conversion is introduced from various T_0 to the true anomaly or θ . In the binary system B-C in LTT 1445, T_0 in Besselian years is converted to the true anomaly by calculating the mean anomaly and using a universal Kepler solver to calculate the value of θ . A similar procedure is done for the outer orbit in LTT 1445, however the mean anomaly is $M = \lambda_{ref} - \omega$. The planets in LTT 1445 have a time of conjunction, so a conversion to the time of periastron is needed first, and then the procedure follows the previous examples. The following expressions were used for this conversion:

$$\theta_{conj} = \frac{\pi}{2} - \omega E_{conj} = 2 \arctan\left[\sqrt{\frac{1-e}{1+e}} \cdot \tan(\theta_{conj}/2)\right] M_{conj} = E_{conj} - e \sin E_{conj} T_0 = T - \frac{P}{2\pi} \cdot M_{conj} \quad (1)$$

Finally, for the LTT 1445 planets, Ω will be set to 0, while the inclination for planet d (as it is non-transiting) will be set to the same inclination as the other two planets under the assumption of coplanarity at the angle of 89° . The N-body code will be fed data from the sky-view, as precise orbits of all the planets and stellar components are available and measured as sky inclinations. Since the stellar components of GJ 667 are poorly constrained, the binary pair AB will be merged into a single object and the orbit of C - (A-B) will be investigated through a study of the parameter space. Noted with (S) in Table 2, the range of values for eccentricity is $\{0.0, 0.2, 0.4, 0.6, 0.8\}$ and for mutual inclination $\{0^\circ, 30^\circ, 60^\circ, 90^\circ\}$ (with this selection, two values are below the critical Kozai-Lidov angle of 39.2° , and two above it). The values of the remaining orbital elements are $\Omega = 0^\circ$, $\omega = 0^\circ$ and $\theta_0 = 90^\circ$. As the confirmed GJ 667 planets are non-transiting, coplanarity is assumed with a mutual inclination of 0 and $\Omega = 0$. Mutual inclination is used in this system as there is no measured sky inclination quantity so the coordinate system is built on mutual inclinations of orbits AB around C and the planets' orbits.

3. Methodology

- exploring dynamical and temperature stability
 - guideline for LTT 1445 is Verier et al 2007 and the main architecture
 - for GJ 667 due to the outer component being a single object, use binary code system and main source
 - link code and github

Table 1: Stellar properties of the trinary sytems GJ 667 and LTT 1445.

Star	LTT 1445				GJ 667			
	$M(M_{\odot})$	$R(R_{\odot})$	$T(K)$	$L_{bol}(L_{\odot})$	$M(M_{\odot})$	$R(R_{\odot})$	$T(K)$	$L_{bol}(L_{\odot})$
A	0.257 ± 0.014	0.268 ± 0.027	3337 ± 150	$0.00794^{(D)}$	-	0.76	4270	$0.173^{(D)}$
B	0.215 ± 0.014	0.236 ± 0.027	-	$0.00596^{(D)}$	-	0.70	3998	$0.113^{(D)}$
C	0.161 ± 0.014	0.197 ± 0.027	-	$0.00368^{(D)}$	0.327 ± 0.008	0.337 ± 0.014	3443^{+75}_{-71}	0.01439 ± 0.00034

Notes. (D) denotes a value calculated or unit-converted. The luminosities of LTT 1445 A,B and C from (Rukdee et al. 2024) are converted in terms of the Sun’s luminosity according to (Prša et al. 2016). The masses, radii and temperatures of LTT 1445 stellar components are adopted from Winters et al. (2019); Rukdee et al. (2024). The masses of components A and B in the GJ 667 system currently have no explicit values, however according to Desgrange et al. (2023) the AB pair has a combined mass of $1.27M_{\odot}$ and can be treated as a single object for N-body integration. Luminosities of stars GJ 667 A and B are adopted Stefan-Boltzmann values, $L/L_{\odot} = (R/R_{\odot})^2(T_{eff}/5772K)^4$, calculated from older component-resolved radii and temperatures Johnson & Wright (1983). The properties of the host, GJ 667 C are adopted from Pineda et al. (2021).

Table 2: Orbital properties of the trinary sytems GJ 667 and LTT 1445.

	LTT 1445		GJ 667
	$B - C$	$A - (B - C)$	$C - (A - B)$
P (yr)	36.2 ± 5.3	549^{+243}_{-136}	-
α (″)	1.159 ± 0.076	-	32.4
a (au)	8.1 ± 0.5	58^{+16}_{-10}	235
e	0.50 ± 0.11	$0.375^{+0.084}_{-0.037}$	$[0.0, 0.2, 0.4, 0.6, 0.8]^{(S)}$
i_{sky} (deg)	89.64 ± 0.13	$88.5^{+1.3}_{-1.4}$	-
i_{mut} (deg)	-	2.88 ± 0.63	$[0.0, 30.0, 60.0, 90.0]^{(S)}$
Ω (deg)	137.63 ± 0.19	135.15 ± 0.28	-
ω (deg)	209 ± 13	256^{+28}_{-21}	-
T_0 (Besselian yr)	2019.2 ± 1.7	-	-
λ_{ref} (deg)	-	210.3 ± 2.1	-

Notes. Orbital data for the binary pair A-B in LTT 1445 was adopted from Winters et al. (2019), while the data for the outer pair is adopted from Zhang et al. (2024). The orbital parameters listed for the outer orbit of LTT 1445 in the table are the posterior median values, but in the evolutions of planetary systems the best-fit values will be used. Since there is no precise estimate of the separate masses of components A and B in GJ 667, their orbit is irrelevant as they will be approximated as a single body. The orbit of component C around the merged binary AB doesn’t have a fully defined orbit (Desgrange et al. 2023), so a range of values for eccentricity and inclination will be included as testing the parameter space for possible stable orbits of this system, denoted by (S).

3.1. Temperature calculation and climate models

- reference and explanation for temperature calculation
- explain methodology of code and the testing for every system

4. Results

5. Discussion

6. Conclusions

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References

Anglada-Escudé, Guillem, Tuomi, Mikko, Gerlach, Enrico, et al. 2013, A&A, 556, A126

Table 3: Properties of exoplanets in trinary sytems GJ 667 and LTT 1445.

	LTT 1445			GJ 667	
	b	c	d	b	c
$M(M_{\oplus})$	2.32 ± 0.25	1.00 ± 0.19	-	-	-
$M \sin i(M_{\oplus})$	-	-	2.72 ± 0.75	5.6 ± 0.3	4.1 ± 0.6
$R(R_{\oplus})$	1.43 ± 0.09	$1.07^{+0.10}_{-0.07}$	-	-	-
$a (au)$	0.03807	0.02656	0.09	0.050431 ± 0.000004	0.12501 ± 0.00009
e	$0.18^{+0.07}_{-0.06}$	$0.05^{+0.05}_{-0.03}$	$0.26^{+0.14}_{-0.13}$	0.15 ± 0.05	0.27 ± 0.1
$i_{sky} (deg)$	$88.95^{+0.64}_{-0.57}$	$86.5^{+0.39}_{-0.47}$	-	-	-
$i_{mut} (deg)$	-	-	-	-	-
$\Omega (deg)$	-	-	-	-	-
$\omega (deg)$	$243.5^{+14.9(D)}_{-17.8}$	$162.7^{+10.3(D)}_{-108.3}$	$50.4^{+49.8(D)}_{-33.8}$	12 ± 20	140 ± 20
$T_0 (BJD)$	$2458423.426378^{+0.000299}_{-0.000304}$	$2457005.358765 \pm 0.000003$	$2457000.04618^{+0.00085}_{-0.00086}$	2454446.1 ± 0.3	2454462 ± 2

Notes. Orbital information and the parameters of planets b, c and d are adapted from Lavie et al. (2023), with the radius of planet c from Pass et al. (2023) due to higher precision. The values of orbital parameters are taken as eccentric posteriors from Lavie et al. (2023), as eccentricity is a key component in habitability. For GJ 667 the values are adopted from Robertson & Mahadevan (2014).

- Anglada-Escudé, G., Arriagada, P., Vogt, S. S., et al. 2012, *The Astrophysical Journal Letters*, 751
- Bennett, K. A., Sing, D. K., Stevenson, K. B., et al. 2025, *The Astronomical Journal*, 169, 111
- Buseti, F., Beust, H., & Harley, C. 2018, *A&A*
- Delfosse, X., Bonfils, X., Forveille, T., et al. 2013, *A&A*, 553, A8
- Desgrange, C., Milli, J., Chauvin, G., et al. 2023, *A&A*, 680
- Eggl, S., Georgakarakos, N., & Pilat-Lohinger, E. 2020, *Galaxies*, 8
- F. Feroz, M. P. H. 2014, *Monthly Notices of the Royal Astronomical Society*, 437, 3540–
- Haghighipour, N. & Kaltenegger, L. 2013, *The Astrophysical Journal*, 777, 166
- Holman, M. J. & Wiegert, P. A. 1999, *The Astronomical Journal*, 117, 621
- Johnson, H. M. & Wright, C. D. 1983, *The Astrophysical Journal Supplement Series*, 53, 643
- Kaltenegger, L. & Haghighipour, N. 2013, *The Astrophysical Journal*, 777, 165
- Kozai, Y. 1962, *The Astronomical Journal*, 67, 591
- Lavie, B., Bouchy, F., Lovis, C., et al. 2023, *A&A*, 673, A69
- Lidov, M. L. 1962, *Planetary and Space Science*, 9, 719
- Makarov, V. V. & Berghea, C. 2013, *The Astrophysical Journal*, 780
- Michel, K.-U. & Mugrauer, M. 2021, *Frontiers in Astronomy and Space Sciences*, 8, 624907
- Müller, T. W. A. & Haghighipour, N. 2014, *The Astrophysical Journal*, 782
- NASA Exoplanet Archive. 2026, Exoplanet and Candidate Statistics, https://exoplanetarchive.ipac.caltech.edu/docs/counts_detail.html, accessed: 2026-06-19
- Pass, E. K., Winters, J. G., Charbonneau, D., et al. 2023, *The Astronomical Journal*, 166, 171
- Pineda, J. S., Youngblood, A., & France, K. 2021, *The Astrophysical Journal*, 918
- Prša, A., Harmanec, P., Torres, G., et al. 2016, *The Astronomical Journal*, 152
- Raghavan, D., Henry, T. J., Mason, B. D., et al. 2006, *The Astrophysical Journal*, 646
- Robertson, P. & Mahadevan, S. 2014, *The Astrophysical Journal Letters*, 793, L24
- Roell, T., Neuhauser, R., Seifahrt, A., & Mugrauer, M. 2012, *A&A*, 542, A92
- Rukdee, S., Buchner, J., Burwitz, V., et al. 2024, *Astronomy & Astrophysics*, 687
- Schlagenhauf, S., Mugrauer, M., Ginski, C., et al. 2024, *Monthly Notices of the Royal Astronomical Society*, 529, 4768
- Sloane, S. A., Guinan, E. F., & Engle, S. G. 2023, *Research Notes of the AAS*, 7
- Verrier, P. E. & Evans, N. W. 2007, *Monthly Notices of the Royal Astronomical Society*
- Winters, J. G., Cloutier, R., Medina, A. A., et al. 2022, *The Astronomical Journal*, 163, 168
- Winters, J. G., Medina, A. A., Irwin, J. M., et al. 2019, *The Astronomical Journal*, 158, 152
- Wolszczan, A. & Frail, D. A. 1992, *Nature*, 355, 145
- Zhang, J., Weiss, L. M., Huber, D., et al. 2024, *The Astronomical Journal*, 167

120 **Appendix A: Additional material**

Appendix text here.